



P37

Absorb and Beware: Patch-Level Amendment Specifications for the Existing UDOS Paper Corpus (P0–P34) — A v7.7.1 Revision Sheet, Not a Direct Rewrite

对 UDOS 论文卷的吸收与警惕：修订规格书



Absorb and Beware: Patch-Level Amendment Specifications for the Existing UDOS Paper Corpus (P0–P34) — A v7.7.1 Revision Sheet, Not a Direct Rewrite

Working Title (EN): Absorb and Beware: Patch-Level Amendment Specifications for the Existing UDOS Paper Corpus (P0–P34) — A v7.7.1 Revision Sheet, Not a Direct Rewrite

Supplement volume P37 • turning 2026 first-half external frontier into "supplementary paragraph specifications directly mergeable into the existing paper corpus" • UDOS v7.7.1 • Fang Wenxin • 2026-09-21

One sentence first: this paper does not change one word of P0–P34, it only produces 7 "**supplementary paragraph specifications**"—4 to absorb, 3 to beware; each written to the granularity "paste into which paper, insert at which section, attach which draft, cite which evidence, mark which status." All external

numbers strictly follow three states (✅ existence verified / 🟡 paper self-reported • not independently rechecked / 🟠 no evidence found), all new architectures, new fields, new connections uniformly marked **design proposal** • **pending implementation**.

Evidence caliber (stated once for the whole paper): UDOS v7.7.1, 2026-09-21, Fang Wenxin. This volume is a **revision specification sheet, not directly changing old papers**; UDOS existing paper corpus is P0–P15 (v7.5.0) and P16–P34 (v7.6.0). External papers/standards/industry sources' existence and number three states **uniformly per EVIDENCE_LEDGER_v771.md**; all ledger 🟡 items, this paper only treats as "directional corroboration," not entering UDOS measured ledger's verified column, all ledger 🟠 items not cited. Verification date 2026-09-21.

1 One-Sentence Conclusion and Structured Abstract

One-sentence conclusion: the external frontier's correct action toward UDOS's existing paper corpus is not "overturn and rewrite," but **4 absorptions adding dimensions, 3 bewarenesses adding guardrails**—each hung back to the original paper as a "supplementary paragraph specification," by default not changing old-version semantics, merged item by item during the v7.7.x release window.

Structured Abstract (background problem → method/argument → evidence → contribution):

- **Background problem:** UDOS existing paper corpus (P0–P15 v7.5.0, P16–P34 v7.6.0) has closed the main proposition "structural priors set capability ceilings"; but a batch of 2026 first-half external works (Warp-Cortex, AgentCollabBench, MCP/A2A, The Last AI Built by Humans, Cost of Consensus, Phase Transition for Budgeted Multi-Agent Synergy, Verification Capacity, CACM column) from seven directions—"memory complexity, collaboration topology, cross-platform handoff, RSI roadmap, homogeneous common-cause faults, budgeted saturation, verification ceiling, tactical/strategic recursion"—both filled dimensions our argument lacked and poked places our narrative might inflate. Directly changing old papers would break closed citation relationships and registered evidence grades.
- **Method/argument:** this paper adopts "**incremental patches**" rather than "rewrite" strategy—for each external finding, producing a five-element supplementary specification (target paper / insertion position / supplementary body draft / evidence citation / status mark), drafts concrete enough to paste directly; and for each giving "how by default not to break old-version semantics" compatibility notes and [RESULT NEEDED] pending measurement items.
- **Evidence:** 7 specifications' external sources all registered per EVIDENCE_LEDGER_v771.md three states—among them ✅ existence verified 4 (The Last AI Built by Humans, MCP/A2A, Phase Transition, Verification Capacity, CACM column all existence confirmed but 细分 claims/numbers mostly retellings), 🟡 paper self-reported not independently rechecked 3 (Warp-Cortex, AgentCollabBench, Cost of Consensus); this paper endorses no vendor/paper.
- **Contribution:** seven paste-ready supplementary specifications hung back to P0/P1/P2/P9/P10/P14; a downgraded citation discipline "external conclusions only as directional hypotheses, not entering verified ledger"; and four falsifiable/rollback conditions for post-hoc judging whether these absorptions stand.

2 What This Paper Is: Supplementary Paragraph Specifications, Not Changing Old Papers

UDOS existing paper corpus divided into two generations:

- **v7.5.0 master:** P0–P15 ("Spatial Structure Itself Is Intelligence," "Million-Scale Extrapolation of Layered Hybrid Topology," "BFT-lite Stop Decision," "Quality Saturation Law," "TransferBundle," "The Convergence of RSI," etc.).
- **v7.6.0 master-outline volume:** P16–P34 (eight structural change specifications and compatibility paths, representative P33 master outline).

This paper **does not modify any of the above's body text**. It only produces one artifact—**supplementary paragraph specifications**. Each specification fixed with five elements:

1. **Target paper:** which old-volume paper this supplement should finally be pasted into (P0 / P1 / P2 / P9 / P10 / P14).
2. **Insertion position:** after which section/paragraph of that paper to insert (precise to section number or subsection name).
3. **Supplementary body draft:** a directly paste-ready Chinese body (this section a draft, not formal final).
4. **Evidence citation:** which external source to cite, and its three-state grade.
5. **Status mark:** uniformly marked "**suggestion • pending merge into v7.7.x**", and marked "absorption item" or "bewareness item."

Discipline restated: once an old volume closes, its internal citation relationships, evidence grades, figure numbers are stable assets. External new findings can only attach as "patch paragraphs," **must not retroactively change conclusions already drawn in old paragraphs**; if some patch conflicts with an old conclusion, should explicitly declare the conflict and pending review in the patch, rather than quietly overwriting.

3 Absorption Items (Four to Absorb into UDOS)

The following four written per five elements. Each ending with "**suggestion • pending merge into v7.7.x**" status.

Absorption ①: Merge into P1 Topology Extrapolation — Warp-Cortex's Memory Complexity Conclusion

[Five elements]

- **Target paper:** P1 "Million-Scale Extrapolation of Layered Hybrid Topology Coordination Complexity."
- **Insertion position:** in P1 Section 3 "Million-Scale Extrapolation Analysis," after the two paragraphs "fan-in complexity $\$O(N)\$$ to $O(\log N)\$$ " and "round complexity," add a subsection "3.x Memory Complexity Dimension."
- **Supplementary body draft:**

Supplementary dimension—memory complexity. P1's earlier argument for layered tree advantages concentrated on the two complexity axes fan-in and rounds. External work Warp-Cortex (arXiv: 2601.01298, 🟡 single-author preprint, not independently rechecked) proposes two comparable structural conclusions: first Singleton Weight Sharing, all Agents sharing the same weights, pressing weight memory from $O(N \cdot L)$ linear in Agent count to approximately $O(1)$ weights; second Topological Synapse, using sparse topological connections to press context/synapse memory to $O(N \cdot k)$ (k fan-out degree). This suggests: layered trees' advantage over flat fully connected, besides fan-in $O(\log N)$ and round decline, should add a third axis—memory complexity. Must emphasize: Warp-Cortex is a single-author, no-institutional-attribution extremely thin preprint, its "single RTX 4090, 100 concurrent only 2.2GB VRAM, theoretical capacity >1000" all self-reported, this paper only borrows its structural claim "memory is also an independent complexity axis," not adopting any specific VRAM/concurrency number; its abstract also does not contain "98% context compression no semantic loss," hence not citing that claim. (suggestion • pending merge into v7.7.x)

- **Evidence citation:** Warp-Cortex, An Asynchronous, Memory-Efficient Architecture for Million-Agent Cognitive Scaling on Consumer Hardware, arXiv:2601.01298v1 (2026-01-03, single author Jorge L. Ruiz Williams). Three-state: 🟡 paper self-reported • not independently rechecked; and high-risk self-report, must **downgrade citation**—only cite structural claim, not performance numbers.
- **Status mark:** Absorption ① | **suggestion • pending merge into v7.7.x.**

Compatibility note: P1's old body fan-in/rounds $O(\log N)$, $O(N)$ conclusions not one line changed; this item only appends "third complexity axis" comparison discussion after it, not replacing any verified extrapolation conclusion. [RESULT NEEDED: in UDOS layered tree vs flat fully connected controlled comparison, actually measuring the two's weight memory and context memory scaling curves with Agent count N , verifying whether "memory complexity" is genuinely an independent third axis; if measurement not significant, this item downgrades to a side note.]

Absorption ②: Merge into P0b F5 Cross-Substrate Collapse — AgentCollabBench's "Collaboration Topology" Independent Substrate

[Five elements]

- **Target paper:** P0b "Structure Plus Scale, Everything Can Be Scaling Laws" F5 item "Cross-Domain Collapse Prediction."
- **Insertion position:** after P0b F5 body "structure sets curve shape" assertion, add a corroboration registration paragraph.
- **Supplementary body draft:**

Corroboration registration—collaboration topology as independent substrate dimension. P0b F5's core assertion is "cross-domain collapse: structure sets curve shape, scale only sets height." External work AgentCollabBench (arXiv:2605.08647, 🟡 based on 900 human-verified tasks, covering software engineering/DevOps/data engineering) independently found: multi-Agent system reliability is essentially a structural problem, simply enlarging the underlying model's intelligence cannot replace collaboration architecture design. This forms cross-domain cross-validation with F5's "structure sets curve shape"—suggesting explicitly listing "collaboration topology (who connects to whom, how handoffs happen)" as an independent substrate dimension alongside "model scale": with collaboration topology fixed, enlarging the model only raises "height"; to change "curve shape," must change topology itself. Must note: this paper currently an arXiv preprint, "ICML 2026 acceptance" not first-hand

verified, this paper only adopts its "900 human-verified tasks" and "reliability is structural" directional conclusions, not adopting any benchmark scores. (suggestion • pending merge into v7.7.x)

- **Evidence citation:** AgentCollabBench, Diagnosing When Good Agents Make Bad Collaborators, arXiv:2605.08647v1 (2026-05-09). Three-state: 🟡 paper self-reported • not independently rechecked; "ICML 2026" not verified, cite as arXiv preprint.
- **Status mark:** Absorption ② | suggestion • pending merge into v7.7.x.

Compatibility note: F5's old assertion "structure sets curve shape" unchanged; this item only explicitly writes "collaboration topology" dimension into the dimension list, and registers an external independent finding as corroboration, not changing F5's prediction caliber. [RESULT NEEDED: inside UDOS doing "same model, different collaboration topology" comparison—fixing underlying model, only changing handoff topology, measuring whether task success curves undergo shape (not height) changes, to verify "collaboration topology is independent substrate."]

Absorption ③: Merge into P10 TransferBundle — MCP/A2A Standardization as External Extension Direction

[Five elements]

- **Target paper:** P10 "Measurability of TransferBundle Handoff Information Loss."
- **Insertion position:** in P10 final section "Discussion and Future Work," add an "external standardization connection direction."
- **Supplementary body draft:**

Future extension—connecting with external handoff standards. UDOS's TransferBundle currently a self-developed six-field contract: Goal / Context / Done / Todo / Trace / Owner (v7.6.0 additionally adds scope field). In 2026 the industry saw two handoff-related standardization moves: first, MCP on 2026-07-28 released the "largest revision," changing to stateless core—removing protocol-level sessions, round-robin scalable on ordinary HTTP load balancing (✅ existence verified); second, A2A released v1.0, introducing Signed Agent Card (cryptographically signed Agent business card), and on 2026-08-17 merged into Linux Foundation AAIF (✅ existence verified; "150+ supporting organizations" etc. specific numbers not verified, not cited). Accordingly suggesting: TransferBundle need not replace the self-developed six fields, but should plan an external extension interface—mapping the six fields' Owner/Scope to A2A's Signed Agent Card, aligning stateless handoff to MCP's stateless core, thereby achieving cross-platform Agent handoff information standardization and verifiability. This connection a design proposal • pending implementation, current UDOS still takes the internally developed six fields as standard. (suggestion • pending merge into v7.7.x)

- **Evidence citation:** MCP official blog "The 2026-07-28 Specification"; A2A official "A2A Protocol Ships v1.0" (Signed Agent Card); AAIF official announcement (A2A joined 2026-08-17). Three-state: ✅ existence verified; specific ecosystem scale numbers (e.g. "150+ organizations") 🟡 not verified, not cited.
- **Status mark:** Absorption ③ | suggestion • pending merge into v7.7.x.

Compatibility note: P10's self-developed six fields (+v7.6.0 scope) unchanged, still default; external connection only a plan in the "future work" subsection, not introducing runtime branches, not breaking P10's existing 26 contract tests. [RESULT NEEDED: designing and verifying TransferBundle six fields ↔ A2A Signed

Absorption ④: Merge into P14 Convergence Criterion — The Last AI Built by Humans' RSI Five-Stage Roadmap

[Five elements]

- **Target paper:** P14 "The Convergence of RSI" ("optimization object five-stage chain").
- **Insertion position:** P14 "optimization object five-stage chain" subsection, inserted as its upper-framework explanation.
- **Supplementary body draft:**

Upper framework—RSI five-stage roadmap. P14 already has the "optimization object five-stage chain," and uses L2–L5 Harness-RSI to characterize self-improvement depth. External work The Last AI Built by Humans: Toward Genuine Recursive Self-Improvement (arXiv:2609.11873v2,  existence verified) gives an alignable RSI five-stage roadmap: ① improve execution autonomy → ② improve strategy autonomy → ③ experience acquisition autonomy → ④ environment adaptation autonomy → ⑤ recursive meta-improvement. Suggesting mapping P14's L2–L5 Harness-RSI to the roadmap's middle stages—L2 (modify Harness/memory) ≈ stage ① execution autonomy transitioning to stage ② strategy autonomy; L3–L4 ≈ stage ③ experience acquisition, stage ④ environment adaptation; only genuine "successor re-entry closed loop, meta-layer improving improvement itself" corresponds to stage ⑤. This paper also proposes HCI (Headroom-Closed Index), usable as a tool quantifying "how much headroom the current system still lacks to the next stage." This mapping an analysis framework borrowing, not changing P14's original L1–L5 level definitions. (suggestion • pending merge into v7.7.x)

- **Evidence citation:** The Last AI Built by Humans: Toward Genuine Recursive Self-Improvement, arXiv:2609.11873v2 (v1 2026-09-10, v2 2026-09-15). Three-state:  existence verified (five stages and HCI both confirmed in abstract).
- **Status mark:** Absorption ④ | suggestion • pending merge into v7.7.x.

Compatibility note: P14's original L1–L5 definitions and "none called L5" terminology discipline unchanged; this item only adds an external roadmap above it as "alignment coordinates," HCI introduced as optional quantification tool, by default not computed. [RESULT NEEDED: using HCI to do one headroom measurement of UDOS's current Harness-RSI system, landing at which of five stages, and giving a list of "which headrooms to close to enter the next stage."]

4 Bewareness Items (Three to Beware / Write into Threat Analysis)

The following three likewise written per five elements; the difference being they are not "adding dimensions," but "adding guardrails"—preventing UDOS from stepping on pitfalls in external narratives.

Bewareness ①: Merge into P2 Common-Cause Fault Threat — Cost of Consensus's Homogeneous QA

[Five elements]

- **Target paper:** P2 "BFT-lite for Multi-Agent Stop Decisions" "threat analysis / common-cause faults" subsection.
- **Insertion position:** in P2's common-mode failure case list, add a "homogeneous QA" case.
- **Supplementary body draft:**

Newly registered common-cause fault case—homogeneous QA. P2's BFT-lite uses equal-weight $2f+1$ and view change to mechanistically avoid "minority obeying majority voted by the same bias," but threat analysis must still explicitly register a real common-cause fault: homogeneous QA / homogeneous multi-Agent debate. External work The Cost of Consensus (arXiv:2605.00914, ; note it an arXiv preprint, not "ACM CAIS 2026-05") reports: in unguided homogeneous multi-Agent debate, there exists sycophantic compliance leading to modal adoption convergence, and homogeneous debate relative to isolated self-correction costs about $2.1\text{--}3.4\times$ more tokens; its two specific numbers "modal adoption rate 85.5%," "consensus collapse oracle gap 32.3pp" not verified in public abstract, this paper only adopts the directional conclusion—"isolated self-correction beats unguided homogeneous debate." Must explicitly register: although BFT-lite in voting mechanism does not rely on homogenization, as long as a group of Agents share the same base, same prompt, same bias, their "independent" judgments on the same problem are actually correlated, still constituting a common-cause fault; UDOS's multi-Agent design must explicitly defend on "member heterogeneity," not only rely on BFT voting structure. (suggestion • pending merge into v7.7.x)

- **Evidence citation:** The Cost of Consensus: Isolated Self-Correction Prevails Over Unguided Homogeneous Multi-Agent Debate, arXiv:2605.00914. Three-state:  paper self-reported • not independently rechecked; venue arXiv preprint; 85.5% / 32.3pp not verified in abstract, only cite direction (isolated self-correction wins, homogeneous debate costs $2.1\text{--}3.4\times$ tokens).
- **Status mark:** Bewareness ① | suggestion • pending merge into v7.7.x.

Compatibility note: P2's $2f+1$ quorum, view change, ten property tests all unchanged; this item only adds one common-cause fault case registration in threat analysis, and requires design docs explicitly state "whether members genuinely heterogeneous," not changing the voting algorithm itself. [RESULT NEEDED: in UDOS fault injection framework adding a group of "homogeneous members" experiments—letting $2f+1$ members share the same base and bias, measuring voting results' consistent adoption rate of the same error, quantifying this common-cause fault's exposure.]

Bewareness ②: Merge into P9 Quality Saturation Law — Budgeted Saturation + Verification Capacity Ceiling

[Five elements]

- **Target paper:** P9 "Agent Legion Quality Saturation Law and Homogeneous Headcount Expansion Failure Boundary" saturation analysis subsection.
- **Insertion position:** after P9's "homogeneous copies saturate no matter how many added" ($\text{mse}(k)=a+b/k$) discussion, add a subsection "9.x Budgeted Saturation and Verification Capacity Ceiling."
- **Supplementary body draft:**

Two new dimensions of the saturation law. P9 already used homogeneous copies to prove "adding people saturates" (cpu-proto). 2026 external works strengthened this bewareness from two directions: first, Phase Transition for Budgeted Multi-Agent Synergy (arXiv:2601.17311,  existence verified) uses fixed-point analysis to prove, under given budget/strong baseline, multi-Agent synergy exhibits a sharp "amplification–collapse" phase transition—weak signals after a parameter scalar exceeds threshold not amplified into useful fixed points but washed to random, i.e. on strong baselines even saturation or negative returns. Second, industry experience article "Verification Capacity as the Agent Quality Ceiling" (AgentPatterns.ai, 2026-08-09,  existence verified, engineering blog not peer-reviewed) proposes the verification capacity ceiling: when "generation (change arrival) speed faster than verification (cleanup) speed," per Little's Law queues pile, delivery quality degrades to the slowest verifier's speed in the chain; its default countermeasure reduce generation rate / WIP=1. Accordingly suggesting: P9's saturation analysis expands from "homogeneous headcount" one dimension to two—
 ① budget/strong baseline dimension (adding people on strong baselines may negative-return); ② verification capacity ceiling dimension (however many people, as long as verification throughput cannot keep up, quality still stuck). This directly constrains UDOS: before expanding Agents, first ask whether verification throughput suffices. (suggestion • pending merge into v7.7.x)

- **Evidence citation:** Phase Transition for Budgeted Multi-Agent Synergy, arXiv:2601.17311v1 ( existence verified, phase transition theory); AgentPatterns.ai "Verification Capacity Saturation: Three Levers, One Default" (2026-08-09, Addy Osmani;  existence confirmed,  industry experience not peer-reviewed).
- **Status mark:** Bewareness ② | **suggestion • pending merge into v7.7.x.**

Compatibility note: P9's old $\text{mse}(k)=a+b/k$ saturation curve and cpu-proto conclusion unchanged; this item only appends two "why saturated/when negative-return" explanatory dimensions after it, not replacing the registered saturation law.
 [RESULT NEEDED: in UDOS separating "generation throughput" and "verification throughput" two knobs, doing WIP/generation rate scans, measuring whether quality is stuck on the verification side, and locating the Little's Law queue saturation point.]

Bewareness ③: Merge into P14 Against RSI Narrative Inflation — Tactical vs Strategic Recursion

[Five elements]

- **Target paper:** P14 "The Convergence of RSI" (narrative boundary / terminology discipline part).
- **Insertion position:** at P14's discussion of "whether RSI has arrived," add a concept distinction and narrative red line.
- **Supplementary body draft:**

Narrative red line—distinguishing "development speed acceleration" from "architecture frontier breakthrough." To prevent RSI narrative inflation, P14 must separate two conflated concepts: tactical recursion (compressing development speed within the existing capability frontier—i.e. "using AI to make AI's own development pipeline faster") and strategic recursion (redefining "what is worth optimizing," breaking through the existing capability frontier itself). BLOG@CACM column (2026-07-06, journalist Logan Kugler,  existence verified but a news column, not peer-reviewed) citing MIT CSAIL view points out: data collection (needing humans to identify the model's own blind spots) and physical world observability are RSI's two fundamental bottlenecks—meaning, even if development speed recursively compressed, as long as "what to optimize" and "where model blind spots are" still rely on

human identification, the system stays in tactical recursion, not entering strategic recursion. Accordingly suggesting: P14 in citing any "AI self-improvement/self-creation" narrative (including vendor self-statements), must mark whether it belongs to development speed acceleration or architecture frontier breakthrough; the former engineering efficiency improvement, the latter approaching genuine RSI, the two not to be conflated. (suggestion • pending merge into v7.7.x)

- **Evidence citation:** BLOG@CACM, Is Recursive Self-Improvement Really Here? (Logan Kugler, 2026-07-06). Three-state:  existence verified, but news column not research paper; "tactical/strategic recursion" dichotomy and MIT CSAIL view column retellings, not rechecked from first-hand source.
- **Status mark:** Bewareness ③ | suggestion • pending merge into v7.7.x.

Compatibility note: P14's original L1–L5 levels and "none called L5" discipline unchanged; this item only adds a "narrative marking rule"—whenever citing external RSI claims, first applying "tactical vs strategic" labels, not changing existing level definitions. [RESULT NEEDED: building an "RSI narrative location table," marking 2026 each self-statement (e.g. model self-report "instrumental in creating itself," R&D automation index) tactical/strategic one by one, counting the proportion genuinely touching strategic recursion, preventing narrative inflation.]

5 Compatibility and Pending Items Master Table

Table 1 Seven supplementary specifications quick reference (all "design proposal • pending implementation / suggestion • pending merge into v7.7.x")

#	Type	Target paper	Insertion position	External source (three-state)	Default compatibility (not breaking old semantics)	Pending measurement [RESULT NEEDED]
Absorption ①	absorb	P1 million extrapolation	Section 3 add "memory complexity" subsection	Warp-Cortex arXiv: 2601.01298 (🟡 downgrade)	fan-in/rounds $O(\log N)$ conclusion unchanged, only append third-axis comparison	measure weight/context memory scaling with N
Absorption ②	absorb	P0b F5 cross-substrate collapse	after F5 assertion add corroboration	AgentCollabBench arXiv:2605.08647 (🟡)	"structure sets curve shape" unchanged, only list collaboration topology as independent dimension	same model change topology, measure whether curve shape changes
Absorption ③	absorb	P10 TransferBundle	final section add external standardization direction	MCP 2026-07-28 / A2A v1.0 (✅ existence)	self-developed six fields default, connection only "future work"	six fields ↔ A2A Agent Card mapping fidelity
Absorption ④	absorb	P14 convergence criterion	on five-stage chain add roadmap framework	The Last AI Built by Humans arXiv: 2609.11873 (✅)	L1–L5 definitions unchanged, only align coordinates	use HCI to measure UDOS current stage headroom
Bewareness ①	beware	P2 common-cause fault	threat analysis add "homogeneous QA" case	Cost of Consensus arXiv:2605.00914 (🟡 direction)	$\$2f+1\$/\text{view}$ change unchanged, only register case	homogeneous member fault injection consistent adoption rate
Bewareness ②	beware	P9 saturation law	saturation analysis add two dimensions	Phase Transition arXiv:2601.17311 (✅) + AgentPatterns.ai (✅🟡)	$\text{\$}\{mse\}(k)=a+b/k$ unchanged, only add explanatory dimensions	generation/verification throughput separation scan, locate queue saturation
Bewareness ③	beware	P14 narrative boundary	add "tactical vs strategic" red line	BLOG@CACM 2026-07-06 (✅ column)	L1–L5 and "none L5" unchanged, only add marking rule	RSI narrative location table, count genuine strategic recursion proportion

Compatibility master principles (following v7.6.0 P33's three sentences, this item follows):

1. **Default is old version:** all seven supplements are "appended paragraphs/registered cases/marketing rules," not replacing, not retroactively changing old-volume closed conclusions; not actively opening any new behavior, old-version semantics verbatim preserved.
 2. **Switches explicit:** all runtime-involving items (e.g. HCI measurement, generation/verification throughput separation, homogeneous fault injection) are new optional experiment switches, default off, before opening not producing new attack surface.
 3. **Old tests not regressing:** P2's ten property tests, P10's twenty-six contract tests in merging into v7.7.x fully regressed, ensuring patches don't break existing.
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6 Dross, Failure Boundaries and Falsifiable Conditions

6.1 This Specification Sheet's Own Dross and Boundaries

- **External paper conclusions cannot directly serve as UDOS measured gains.** All seven are "directional corroboration/framework borrowing/threat registration," not one a number UDOS itself ran. Reading "Warp-Cortex presses to O(1) weights" as "UDOS already can run thousand Agents on one card" is this volume's biggest misuse.
- **Self-reported numbers only as directional hypotheses, not entering verified ledger.** Warp-Cortex (🟡 single-author extremely thin preprint), AgentCollabBench (🟡 preprint), Cost of Consensus (🟡, and 85.5%/32.3pp not verified in abstract) three, this paper uniformly only cites "structural claim/direction," **not citing specific performance numbers, not entering UDOS evidence ledger's verified column.**
- **Standards/columns' "existence" does not equal "claims verified."** MCP/A2A, Phase Transition, AgentPatterns.ai, BLOG@CACM all existence confirmed; but "150+ organizations," "tactical/strategic recursion," "MIT CSAIL view" etc. 细分 claims still retellings/industry experience, cannot cite as theorems.
- **Not endorsing vendors.** All vendor self-statements (e.g. "some model instrumental in creating itself," "R&D automation index") uniformly only as "narrative samples," marked by Bewareness ③'s location table, not written into UDOS capability conclusions.

6.2 Falsifiable / Rollback Conditions

- **F-A (Absorption ① falsifiable):** if in UDOS controlled comparison, layered tree vs flat fully connected weight/context memory scaling **no significant difference**, then "memory complexity as independent third axis" downgrades to a side note, Absorption ① no longer counted in P1's main argument.
- **F-B (Absorption ① downgrade trigger):** if future Warp-Cortex fails third-party reproduction, then Absorption ① immediately downgrades to "falsified direction," removing its structural claim from P1's supplementary paragraph, only retaining "this claim once existed, not reproduced" historical note.
- **F-C (Absorption ② falsifiable):** if fixing underlying model, only changing collaboration topology, UDOS task success curves only undergo height changes, not shape changes, then "collaboration topology is independent substrate" doesn't hold, Absorption ② returns to "corroboration consistent with F5," no longer separately listed as a new dimension.

- **F-D (Bewareness ② boundary):** if in UDOS measurement, after verification throughput maxed expanding Agents still no quality gain (i.e. bottleneck not on verification side), then "verification capacity ceiling" does not constitute a main constraint on UDOS, Bewareness ② downgrades to general engineering advice.

7 Conclusion

v7.7.1's correct posture toward the external frontier is **"paste patches, don't move old walls"**: four absorptions (memory third axis, collaboration topology, cross-platform handoff, RSI five-stage roadmap) hung back to P1/P0b/P10/P14 respectively, adding dimensions the closed main propositions didn't cover then; three bewarenesses (homogeneous common-cause faults, budgeted saturation and verification ceiling, tactical/strategic recursion narrative inflation) hung back to P2/P9/P14 respectively, adding guardrails to already sufficient conclusions. All external numbers guard three states, all new connections marked "design proposal • pending implementation," all drafts marked "suggestion • pending merge into v7.7.x"—**this is a seven-segment paste-ready construction list, not an already delivered results report**; each segment's stay-or-go left to §6's falsifiable conditions and v7.7.x's measurements to score.

References

UDOS existing paper corpus (this series, not modified; evidence grades per each paper)

1. v7.5.0 master: P0 "Spatial Structure Itself Is Intelligence," P0b "Structure Plus Scale, Everything Can Be Scaling Laws" (F5 cross-domain collapse), P1 "Million-Scale Extrapolation of Layered Hybrid Topology Coordination Complexity," P2 "BFT-lite for Multi-Agent Stop Decisions," P9 "Agent Legion Quality Saturation Law," P10 "Measurability of TransferBundle Handoff Information Loss," P14 "The Convergence of RSI."
2. v7.6.0 master-outline volume: P16–P34 (representative P33 "v7.6.0 Architecture Master Outline: Eight Structural Change Specifications and Compatibility Paths").

External dialogue objects (three states per EVIDENCE_LEDGER_v771.md)

1. Warp-Cortex. An Asynchronous, Memory-Efficient Architecture for Million-Agent Cognitive Scaling on Consumer Hardware. arXiv:2601.01298v1 (2026-01-03, single author). 🟡 self-reported • not independently rechecked, downgrade citation.
2. AgentCollabBench. Diagnosing When Good Agents Make Bad Collaborators. arXiv:2605.08647v1 (2026-05-09). 🟡 preprint, "ICML 2026" not verified.
3. MCP official blog "The 2026-07-28 Specification" (stateless core); A2A "A2A Protocol Ships v1.0" (Signed Agent Card); AAIIF official announcement (A2A joined 2026-08-17). ✅ existence; ecosystem scale numbers not verified.
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Evidence Discipline and Reproduction Notes

- This paper is a **revision specification sheet**, no new UDOS experiments; all seven are **design proposals** • **pending implementation**, gain slots [RESULT NEEDED] , forbidden to write as measured.
- External sources' existence, numbers, three-state grades uniformly per EVIDENCE_LEDGER_v771.md (verified 2026-09-21);  items only as directional corroboration, not entering verified ledger;  items (not cited in this volume) uniformly not written.
- This paper **does not modify any P0–P34 old paragraph**; the merge action occurs in the v7.7.x release window, at merge needing P2/P10 full regression.